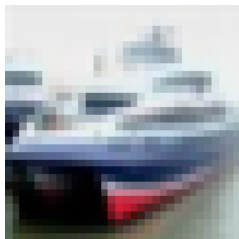
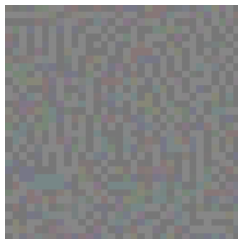


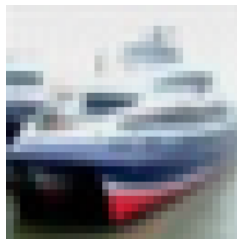
Adversarial robustness: a tiny worst-case change flips the label



clean: "ship" ✓



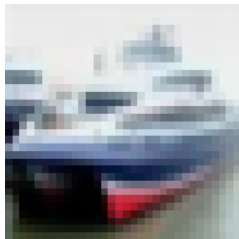
perturbation ($\times 8$)



adversarial: "car" ✗

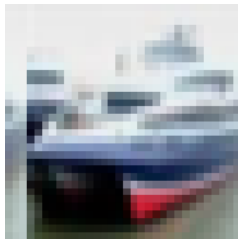
The two images differ by an l_∞ change of size $1/255$, imperceptible to a human, yet the model's label flips.

Invariance: shifting the image keeps the label



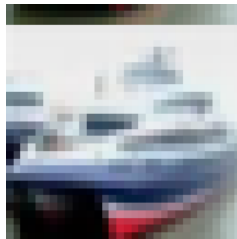
"ship" ✓

original



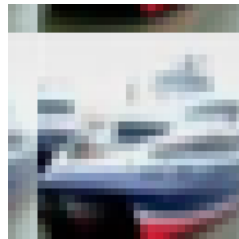
"ship" ✓

shift (3,0)



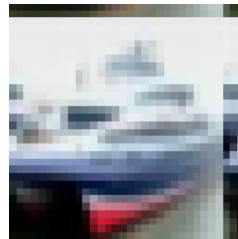
"ship" ✓

shift (0,3)



"ship" ✓

shift (4,4)



"ship" ✓

shift (-3,2)